BME205 – Fundamentals of Biomedical Engineering

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With Exam Question Callouts

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1 Homeostasis & Negative Feedback

Whole-body homeostasis maintains stable internal conditions using a negative feedback loop. Key variables that the body regulates include: body temperature, blood glucose, blood pH, blood pressure, and ion concentrations (e.g., K^+, Na^+, Ca^{2+}). These are called **regulated variables**.

1.1 Main Control Pathway

1.1.1 Control Pathway Components

a. **Reference point (set point)** — the desired value of a regulated variable. Example: body temperature $\approx 37°C$.

b. **Sensor (Receptor)** — detects changes in the regulated variable. Example: thermoreceptors in the skin.

c. **Afferent neurons** — carry sensory information from the sensor to the brain.

d. **Error detector** — compares the measured value to the reference point. If the values differ, an error signal is generated.

e. **Controller** — determines the appropriate response. For temperature: thermoregulatory center in the hypothalamus.

f. **Efferent neurons** — carry control signals from the brain to the effectors.

g. **Effectors** — carry out the response. Examples: sweat glands, blood vessels.

h. **System output** — the response changes the regulated variable. Example: sweating and vasodilation lower body temperature.

Exam Question: (2023 Q1): "Which of these statements best describes negative feedback loops that influence homeostasis?" Answer: **c) Efferent nerves are excited, and the system diminishes a particular physical response.**

1.2 Negative Feedback ★

1.2.1 Key Features of Negative Feedback

a. Negative feedback is a control mechanism in which the **response counteracts the initial change** to restore the system toward its set point

b. The system works to return a variable back toward its set point

c. The response always opposes the initial change (stimulus)

d. Maintains stable internal conditions (homeostasis)

1.2.2 Negative Feedback Loop Steps

a. A stimulus causes a deviation from the set point

b. A receptor senses the change

c. The control center compares the value to the set point

d. An effector produces a response

e. The response reduces the original error

1.2.3 Important Notes

a. Negative feedback does **not eliminate** the stimulus, only reduces it

b. Most physiological regulation uses negative feedback

c. Opposite of positive feedback

2 Cell Biology

Cells are the basic structural and functional units of all living organisms.

2.1 Cell Structure

2.1.1 Cell Membrane ★

The cell membrane (plasma membrane) is a **phospholipid bilayer** that surrounds every cell.

a. Structure: two layers of lipids with hydrophobic tails facing inward and hydrophilic heads facing outward

b. **Selectively permeable** — small nonpolar molecules (O_2, CO_2) pass freely; ions and large molecules(ATP, proteins) need channels, carriers, or pumps

c. Hosts proteins — receptors, channels, enzymes, and pumps are embedded in or attached to the membrane

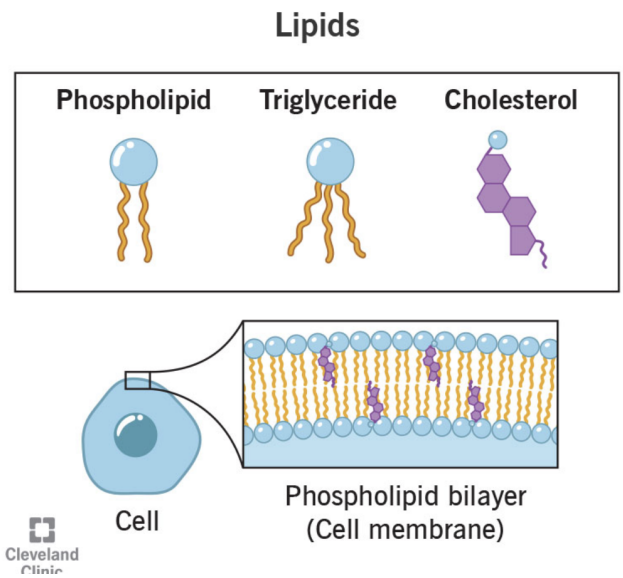


Figure 2: Structure of a phospholipid with hydrophilic head and hydrophobic tails.

Exam Question: (2023 Q4): "Which of the following can't cross a lipid bilayer?" Answer: **a) Ions because they are charged.**

Exam Question: (2023 Q5): "Do steroid molecules need a channel to be transported or diffuse across the cell membrane?" Answer: **b) No, because they are hydrophobic and diffuse freely.**

2.1.2 Key Spatial Terms

a. **Extracellular** — outside the cell. Na^+ and Cl^- are concentrated here.

b. **Intracellular** — inside the cell. K^+ is concentrated here.

c. **Cytoplasm** — gel-like fluid filling the inside of the cell (excluding the nucleus). Contains organelles, dissolved ions, proteins, and nutrients.

2.1.3 Organelles

a. **Nucleus** — contains DNA; the control centre of the cell. Transcription happens here.

b. **Ribosome** — the site of protein synthesis (translation). Made of rRNA and proteins. Can be free in the cytoplasm or attached to the rough ER.

c. **Mitochondria** — the powerhouse of the cell. Produces ATP through oxidative phosphorylation. Neurons have enormous numbers of mitochondria.

d. **Endoplasmic reticulum (ER)**:

- Rough ER — studded with ribosomes; synthesizes and folds proteins
- Smooth ER — no ribosomes; synthesizes lipids and detoxifies substances

e. **Golgi apparatus** — packages, modifies, and sorts proteins for transport (cell's post office)

f. **Vesicles** — small membrane-bound sacs that transport substances within the cell or release contents outside (exocytosis)

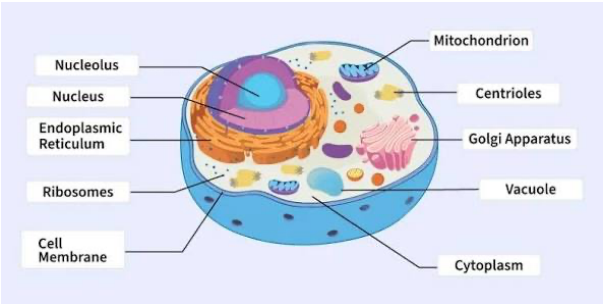


Figure 3: Overview of a typical animal cell showing key organelles.

Exam Question: (2025 MC4): "Which of the following options best describes the function of the ribosome in a cell?" Answer: **c) The ribosomes are like hobby makers using a computer aided design file to create a 3D printed object.**

Exam Question: (2023 Q8): "Which of these statements does NOT apply to ribosomes?" Answer: **c) They are covered by a membrane.**

2.1.4 Hormones

A **hormone** is a chemical messenger produced by a gland or specialized cell and released into the bloodstream. It travels through the blood to act on target cells that have the right receptor.

Important: Hormones are **not RNA**. They are proteins, peptides, or steroids (lipid-based). Hormones are a *result* of gene expression (DNA → mRNA → protein hormone), not part of the process itself.

2.1.5 Tissues

A **tissue** is a group of similar cells that work together to perform a specific function. Cells → tissues → organs → organ systems.

The four primary tissue types:

a. **Epithelial tissue** — covers body surfaces and lines cavities

b. **Connective tissue** — supports, connects, and separates other tissues. Includes bone, cartilage, blood, and the layers wrapping nerves

c. **Muscle tissue** — contracts to produce movement (skeletal, smooth, cardiac)

d. **Nervous tissue** — transmits electrical signals. Made of neurons and glial cells

2.2 Cell Transport ★

2.2.1 Passive Transport (No Energy Required)

Substances move **down** their concentration gradient (high → low).

a. **Simple diffusion** — small nonpolar molecules pass directly through the phospholipid bilayer (e.g., O_2, CO_2)

b. **Facilitated diffusion** — larger or charged molecules move through channels or carriers in the membrane. Still down the gradient, still no energy.

- Channel-mediated — ion channels (e.g., K^+ leak channels)
- Carrier-mediated — a carrier protein binds, changes shape, and releases

c. **Osmosis** — diffusion of water across a selectively permeable membrane

2.2.2 Active Transport (Energy Required) ★

Substances move **against** their concentration gradient (low → high). Requires ATP.

a. **Primary active transport** — a pump directly uses ATP.
Key example: **Na^+/K^+ ATPase pump** — pumps 3 Na^+ out and 2 K^+ in per ATP.

b. **Secondary active transport** — uses the energy stored in an existing ion gradient (created by primary active transport). No direct ATP use, but depends on ATP indirectly.

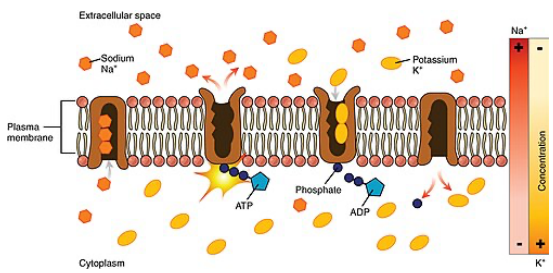


Figure 4: The Na^+/K^+ ATPase pump moves 3 Na^+ out and 2 K^+ in per ATP.

Exam Question: (2024 Q1): "What is the main energy source for the Na^+-K^+ pump in active transport?" Answer: **B) ATP.**

2.2.3 Vesicular Transport

a. **Exocytosis** — vesicles fuse with the cell membrane and release contents outside. Example: neurotransmitter release.

b. **Endocytosis** — the cell membrane wraps around an extracellular substance and pulls it inside.

c. **Cytosis** — general term covering both exocytosis and endocytosis.

Exam Question: (2025 Q13): "Figure 2 displays several molecular transport mechanisms at the neuronal synapse. Label a, b, c, d." Answers: a) Facilitated diffusion, channel mediated; b) Exocytosis; c) Facilitated diffusion, channel mediated; d) Secondary active transport, carrier mediated (reuptake).

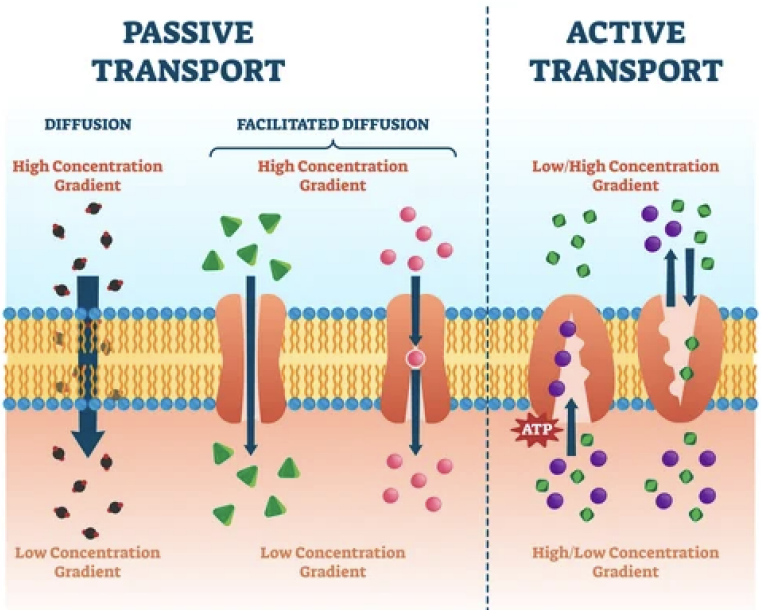


Figure 5: Comparison of passive and active transport mechanisms across the cell membrane.

2.2.4 Flux

Flux is the rate of movement of ions or molecules across a membrane. Net flux depends on:

a. The concentration gradient (chemical driving force)

b. The electrical gradient (for charged particles)

c. The permeability of the membrane to that substance

2.3 Cell Energy ★

2.3.1 ATP

ATP (adenosine triphosphate) is the cell's primary energy currency.

a. Made of adenine + ribose sugar + three phosphate groups

b. Energy is released when the terminal phosphate bond is broken (hydrolysis):

$$\mathrm{ATP} + \mathrm{H_2O} \xrightarrow{\mathrm{ATPase}} \mathrm{ADP} + \mathrm{P}_i + \text{energy}$$

c. ATP is continuously recycled

d. Most ATP is produced by **oxidative phosphorylation** in the mitochondria

2.3.2 Glycolysis and ATP Production Locations ★

a. **Glycolysis** occurs in the **cytosol** (cytoplasm), **NOT** in the mitochondria

b. **Krebs cycle** (citric acid cycle) occurs in the **mitochondrial matrix**

c. **Oxidative phosphorylation** occurs at the **inner mitochondrial membrane** — produces the **most ATP** (this is tested almost every year)

Exam Question: (2025 MC2): "In the game Cytosis, there was a way to get more ATP in a turn if you had an extra glucose token. In real mitochondria, what accounts for the most ATP production?" Answer: **b) Electron transport and oxidative phosphorylation.**

Exam Question: (2024 Q18): "Which process generates the most ATP during cellular respiration?" Answer: **B) Oxidative phosphorylation.**

Exam Question: (2024 Q7/2023 Q7/2025 MC8): "Where are the enzymes for glycolysis located?" Answer: **b) In the cytosol.**

Metabolic Pathways and Their Cellular Locations

Pathway	Location
Glycolysis	Cytosol
Pyruvate Dehydrogenase	Mitochondrial Matrix
Citric Acid (Krebs) Cycle	Mitochondrial Matrix
Electron Transport Chain	Inner Mitochondrial Membrane
Fatty Acid Synthesis	Cytosol
Beta-Oxidation	Mitochondrial Matrix
Pentose Phosphate Pathway	Cytosol

Exam Question: (2023 Q6): "Which of the following options best describes the function of the mitochondria in a cell?" Answer: **a) The mitochondria are like the engine of a car, providing energy to power functions of a cell.** (Note: also accepted b — like fuel in a car.)

Tested EVERY year: Know where each stage of cellular respiration happens and that oxidative phosphorylation produces the most ATP. This is a guaranteed question.

3 DNA ★

DNA (Deoxyribonucleic Acid) stores genetic information.

3.1 DNA Building Blocks

DNA is made of repeating units called **nucleotides**.

3.1.1 Nucleotide Components

Each nucleotide has:

a. a phosphate group

b. a deoxyribose sugar

c. a nitrogenous base

Exam Question: (2024 Q33/34): "What is the basic distinguishing structural unit of DNA?" Answer: **B) Nucleotide.** "What makes up a nucleotide?" Answer: **C) A sugar, a phosphate group, and a nitrogen base.**

3.1.2 The Four DNA Bases

A (Adenine), T (Thymine), C (Cytosine), G (Guanine). The sugar and phosphate form the backbone.

3.2 Base Pairing and Direction

3.2.1 Complementary Base Pairing

a. A pairs with T (2 hydrogen bonds)

b. C pairs with G (3 hydrogen bonds)

Exam Question: (2024 Q25): "Which type of bond is found between base pairs in the DNA structure?" Answer: **C) Hydrogen bond.**

Exam Question: (2024 Q5): "Which RNA nucleobase pairs with adenine in DNA?" Answer: **D) Uracil.**

3.2.2 Antiparallel Strands

The two strands run in opposite directions — this is called **antiparallel**.

a. One strand is 5' → 3'

b. The other is 3' → 5'

3.3 Why DNA Has the Same Width (IMPORTANT NOTE)

3.3.1 Base Size Categories

a. **Purines** (A, G) are large, two-ring bases

b. **Pyrimidines** (C, T) are small, one-ring bases

3.3.2 Why Pairing Keeps Width Constant

Base pairing is always Purine + Pyrimidine. This keeps the distance between the two backbones constant.

3.3.3 DNA Right-Hand Twist ★

DNA forms a **right-handed** double helix. To visualize this:

a. Place the **thumb of your right hand** along the axis of the DNA strand

b. Your **curled fingers** follow the direction of the twist (the base pair steps spiral upward to the right)

c. This is the standard B-form DNA that exists in your body

Exam Question: (2025 MC14): "Which is the best way to accurately describe how the DNA molecule twists?" Answer: **a) The DNA twists properly when you place the thumb of your right hand along the axis of the strand and your curled fingers can easily walk along the base pair steps.**

Exam Question: (2023 Q2): "Which of the following analogies best describes how DNA functions?" Answer: **c) DNA is like a series of buttons that can be undone as required to access only the required information, without undoing all buttons.**

3.4 Gene Expression ★

Gene expression is the process by which information in DNA is used to make a protein. It consists of two steps: transcription and translation.

Protein is made of amino acids. Messenger RNA (mRNA) is USED (not composed) as the template to make proteins.

RNA includes introns, mRNA, rRNA, and tRNA. Introns are parts of RNA that are removed before a protein is made. Proteins, enzymes, and hormones are **not** RNA.

3.5 Mutations

A **mutation** is a permanent change in the DNA sequence.

Point mutation — a change in a single nucleotide:

a. **Silent mutation** — new codon still codes for the same amino acid (no effect)

b. **Missense mutation** — new codon codes for a different amino acid

c. **Nonsense mutation** — new codon becomes a stop codon, producing a truncated protein

3.5.1 Transcription ★

Transcription is the process of making **RNA from DNA**. It occurs in the **nucleus** (eukaryotes) or the cytoplasm (prokaryotes).

The Three Key Players

a. **Template strand** (antisense strand)

- The strand that RNA polymerase actually reads
- Read in the $\mathbf{3'} \rightarrow \mathbf{5'}$ direction
- Is **complementary and antiparallel** to the mRNA produced

b. **Coding strand** (sense strand)

- Has the **same sequence** as the mRNA (with T instead of U)

- Read in the $\mathbf{5'} \rightarrow \mathbf{3'}$ direction to identify codons
- **Not directly read** by RNA polymerase
- Contains **ATG** as the start codon (= AUG in mRNA)

c. **mRNA** (product)

- Synthesized in the $\mathbf{5'} \rightarrow \mathbf{3'}$ direction
- Complementary to the template strand
- Same sequence as the coding strand (with U instead of T)
- Contains **AUG** as the start codon

Direction Rules

Almost everything in molecular biology goes $\mathbf{5'} \rightarrow \mathbf{3'}$. The **only exception** is RNA polymerase reading the template strand in $\mathbf{3'} \rightarrow \mathbf{5'}$, and it does this specifically so it can *build* the mRNA in $5' \rightarrow 3'$.

Process	Direction
DNA replication (new strand synthesis)	$5' \rightarrow 3'$
mRNA synthesis (transcription)	$5' \rightarrow 3'$
Ribosome reading mRNA (translation)	$5' \rightarrow 3'$
RNA polymerase reading template strand	$3' \rightarrow 5'$
Identifying codons on coding strand	$5' \rightarrow 3'$

Worked Example — Identifying the Coding Strand

Given two complementary strands:

```
Strand 1:    5' -- CTA | CGG | TCA | ATA | TAG | GAT | CAT -- 3'
Strand 2:    3' -- GAT | GCC | AGT | TAT | ATC | CTA | GTA -- 5'
```

Step 1: Rewrite each strand in the $\mathbf{5'} \rightarrow \mathbf{3'}$ direction.

- Strand 1 is already $5' \rightarrow 3'$: `CTA CGG TCA ATA TAG GAT CAT`
- Strand 2 reversed to $5' \rightarrow 3'$: `ATG CTA CTA TAT TGA CCG TAG`

Step 2: Look for **ATG** (start codon) at the beginning.

- Strand 1: starts with CTA → **no ATG**
- Strand 2: starts with **ATG** → ✓ **This is the coding strand!**

To identify the coding strand, rewrite both strands $5' \rightarrow 3'$ and look for **ATG**. The strand with ATG in the $5' \rightarrow 3'$ reading frame is the coding strand.

Stages of Transcription: Happens in the nucleus

1. **Initiation**
   - RNA polymerase binds to the **promoter** region on DNA
   - In eukaryotes: requires **transcription factors** (e.g., TFIID binds the TATA box)
   - DNA unwinds to form the **transcription bubble**
2. **Elongation**
   - RNA polymerase reads template strand **3′ → 5′**
   - Builds mRNA **5′ → 3′** by adding complementary ribonucleotides
   - Base pairing: A→U, T→A, G→C, C→G
   - No primer needed (unlike DNA replication)
3. **Termination**
   - Prokaryotes: Rho-dependent or Rho-independent (hairpin loop)
   - Eukaryotes: polyadenylation signal (AAUAAA) triggers cleavage

Post-Transcriptional Modifications (Eukaryotes Only) ★

1. **5′ cap** — 7-methylguanosine cap added to the 5′ end
   - Protects from degradation; aids ribosome recognition
2. **3′ poly-A tail** — 100–200 adenine nucleotides added to 3′ end
   - Protects from degradation; aids nuclear export
3. **Splicing** — removal of **introns**, joining of **exons**
   - Performed by the **spliceosome** (snRNPs)
   - **Introns** = "in the trash" (removed)
   - **Exons** = "expressed" (kept)
   - Alternative splicing: one gene → multiple proteins

Quick Comparison: DNA vs RNA

Feature	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Structure	Double-stranded	Single-stranded
Start codon	ATG	AUG
Location	Nucleus	Nucleus → Cytoplasm

Exam Question: (2025 Q1–Q3): Given two complementary DNA strands, identify the coding strand (has start codon ATG), identify it by the start/stop codons and 5' to 3' direction, then write the mRNA transcription. Answer for Q3: 5'–AUG—AUC—CUA—UAU—UGA—CCG—UAG–3'

Exam Question: (2024 Q44): "During transcription, DNA is copied into:" Answer: **A) mRNA.**

Exam Question: (2023 Q1, 2024 Q1, 2025 Q1–3): DNA → mRNA transcription and identifying coding/template strands appear EVERY year.

3.5.2 Precursor mRNA (pre-mRNA)

Precursor mRNA (pre-mRNA) is the rough draft of RNA produced directly from DNA.

a. **Exons** — useful sections that code for protein (kept)

b. **Introns** — non-coding "junk" sections (removed)

c. Processing: splicing (remove introns), 5' cap, 3' poly-A tail

Codons You Must Memorize ★

There are **3 stop codons** and **1 start codon**.

Start Codon

- **AUG** — codes for **Methionine (Met)**
- In DNA (coding strand): **ATG**

Stop Codons (Translation Termination)

- **UAA** — "**U** **A**re **A**wful"
- **UAG** — "**U** **A**re **G**one"
- **UGA** — "**U** **G**o **A**way"

All three stop codons start with **U**. They do NOT code for any amino acid. When the ribosome reaches one, a **release factor** (not a tRNA) enters the A site and the peptide chain is released.

Quick Summary Table

Codon	Type	Result
AUG	Start codon	Methionine (Met) — begins translation
UAA	Stop codon	Release factor → peptide released
UAG	Stop codon	Release factor → peptide released
UGA	Stop codon	Release factor → peptide released

3.5.3 Types of RNA and Their Roles

RNA type	Role	Analogy
mRNA	Carries instructions to ribosome	Recipe
tRNA	Delivers amino acids during translation	Delivery truck
rRNA	Builds the ribosome structure	Factory machinery
pre-mRNA	Unprocessed mRNA (contains introns)	Rough draft

Exam Question: (2023 Q3): "The best comparison of how tRNA interacts with the ribosome to create a protein is how:" Answer: **a) A wandering catering server is to a plated service meal.**

3.5.4 Translation ★

Translation converts RNA into protein.

Stages:

1. **Initiation** — ribosome assembles around start codon (AUG); first tRNA carrying methionine binds
2. **Elongation** — ribosome moves along mRNA 5' → 3'; tRNA delivers amino acids; peptide bonds form
3. **Termination** — ribosome reaches a stop codon (UAA, UAG, or UGA); polypeptide is released

Exam Question: (2024 Q12): "The termination of protein synthesis occurs when the ribosome encounters:" Answer: **B) A stop codon.**

Exam Question: (2024 Q16): "The start codon AUG codes for which amino acid?" Answer: **A) Methionine.**

Exam Question: (2025 Q4): "Using the mRNA, what is the translated protein sequence?" Answer: **Met–Ile–Leu–Tyr** (STOP is not translated).

Exam Question: (2025 Q5): "Assume more RNA processing and one codon could be an exon or intron. What is the translated protein sequence that results in a longer peptide chain?" Answer: **Met–Ile–Leu–Tyr–Pro** (the UGA stop codon is removed as an intron).

3.5.5 Non-Coding and Non-Template DNA

Non-coding DNA refers to regions that do not code for proteins (introns, regulatory sequences). The non-template strand (coding strand) has the same sequence as mRNA with T instead of U.

3.5.6 Important Clarification

DNA replication is **not part of gene expression**. It copies DNA using DNA as the template: DNA → DNA.

Process	Template
DNA replication	DNA
Transcription	DNA
Translation	mRNA

4 Neuron Structure and Function ★

A **neuron** is a specialized cell that transmits electrical signals for communication in the nervous system.

4.1 Neuron Structure

4.1.1 Main Parts

a. **Dendrites** — receive and integrate incoming signals

b. **Cell body (soma)** — contains the nucleus and integrates signals

c. **Axon** — transmits electrical signals away from the cell body

d. **Axon terminals** — release neurotransmitters

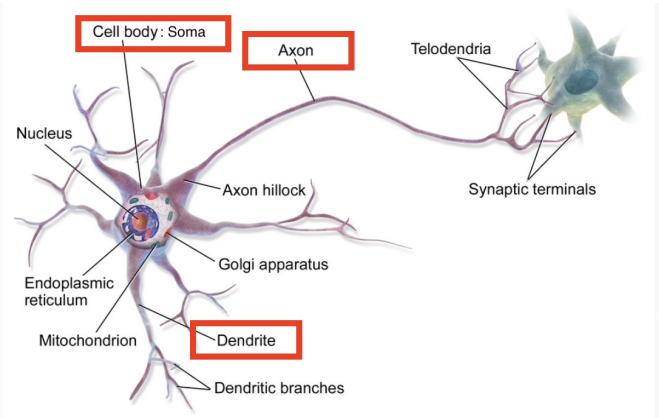


Figure 6: Structure of a neuron showing dendrites, cell body, axon, and axon terminals.

4.2 Key Cellular Components

(Mitochondria, ATPase, Pump, Cell Membrane, Ribosome — as in original notes.)

4.3 Ion Channels ★

4.3.1 Types of Ion Channels

a. **Voltage-gated channels** — open when membrane voltage changes

b. **Ligand-gated channels** — require a chemical ligand to open

c. **Leak channels** — always open; responsible for maintaining RMP

Exam Question: (2024 Q3): "Which type of gated ion channel is crucial for the propagation of action potentials?" Answer: **C) Voltage-gated.**

Exam Question: (2024 Q42): "Which type of ion channel opens or closes in response to a change in membrane potential?" Answer: **B) Voltage-gated channels.**

4.3.2 Resting Membrane Potential (RMP) ★

The **resting membrane potential** (RMP) is approximately **−70mV** — the inside of the neuron is negative relative to the outside.

Ion concentrations (standard values):

Ion	Inside (mM)	Outside (mM)	Direction at rest
K^+	115	2	Wants OUT (chemical)
Na^+	12	150	Wants IN (chemical)
Cl^-	8	150	Wants IN (chemical)

Why K^+ sets the RMP: At rest, K^+ leak channels are always open. K^+ is driven OUT by the chemical force. As K^+ leaves, large negatively-charged proteins trapped inside make the inside more negative. K^+ keeps leaving until the electrical force pulling it back in exactly balances the chemical force pushing it out.

Exam Question: (2024 Q2): "Resting membrane potential is approximately:" Answer: **B) −70 mV.**

The Nernst Equation — calculates the equilibrium potential for a single ion:

$$E_X = \frac{61}{z} \log \frac{[X]_{out}}{[X]_{in}}$$

Exam Question: (2024 Q23): "What does the Nernst equation calculate?" Answer: **B) The equilibrium potential for an ion.**

Exam Question: (2024 Q31): "If $[K^+]_{out} = 5$ mMol/L and $[K^+]_{in} = 150$ mMol/L, what is E_{K^+}?" Answer: **B) −90 mV.** Calculation: $E_{K^+} = \frac{61}{+1} \log \frac{5}{150} = 61 \times (-1.477) = -90.1$ mV.

The Goldman Equation (GHK):

$$V_m = 61 \log \frac{P_{K^+}[K^+_{out}] + P_{Na^+}[Na^+_{out}] + P_{Cl^-}[Cl^-_{in}]}{P_{K^+}[K^+_{in}] + P_{Na^+}[Na^+_{in}] + P_{Cl^-}[Cl^-_{out}]}$$

where $P_{K^+} = 1$, $P_{Na^+} = 0.04$, $P_{Cl^-} = 0.3$.

Exam Question: (2025 Q6): GABA/Cl^-/K^+ calculation using Goldman & Nernst equations with old neuron data. This is the biggest calculation question — appears EVERY year. Know how to calculate V_m and E_{ion} and explain which direction the ion moves.

Na^+/K^+ ATPase Pump:

a. Uses ATP to pump 3 Na^+ out and 2 K^+ in per cycle

b. Maintains the concentration gradients

c. If the pump stops: gradients disappear → RMP collapses toward 0 mV → neuron cannot fire

Hyperkalemia: Elevated extracellular K^+ → reduced K^+ gradient → RMP becomes less negative (depolarized) → initially hyperexcitable, then unexcitable → muscle weakness, cardiac arrhythmias.

Exam Question: (2025 MC9): "For a given ion crossing a cell membrane, the electrical driving force:" Answer: **b) Depends only on the electrical force, regardless of the concentration gradient of the ion.**

Exam Question: (2023 Q9): "For a given molecule crossing a cell membrane, the chemical driving force:" Answer: **b) Depends only on the concentration gradient, regardless of whether or not the molecule is an ion.**

4.4 Action Potential ★

An **action potential** is a brief electrical signal — a rapid, temporary reversal of membrane potential — that travels along a neuron to transmit information.

4.4.1 When Does It Happen?

At rest, the neuron sits at RMP = −70 mV. An action potential only occurs if the membrane is depolarized enough to reach the **threshold potential** (≈ −55 to −60 mV).

All-or-nothing law:

a. If threshold is reached → a full action potential fires

b. If threshold is not reached → nothing happens (no partial AP)

Exam Question: (2025 Q11): "Choose the best analogy for the action potential pattern." Answers A, C, or D accepted with justification. **C (air pistol) is best** — "All or nothing." B (dimmer switch) was NOT accepted because there is nothing "in between."

Exam Question: (2025 MC15): "Suppose the net summation brings the membrane potential to −60 mV. Will an action potential occur?" Answer: **a) No, an action potential will not occur** (threshold is −55 mV, −60 mV is below threshold).

4.4.2 Phases of the Action Potential ★

Phase	Na^+ channels	K^+ channels	Result
Resting (−70 mV)	Closed	Closed	Stable RMP
Depolarization	Open	Closed	V_m rises rapidly (Na^+ rushes in)
Repolarization	Inactivated	Open	V_m falls (K^+ rushes out)
Hyperpolarization	Inactivated	Slowly closing	V_m dips below RMP
Recovery	Closed (reset)	Closed	Returns to RMP

Exam Question: (2023 Q9 / 2024 Q9): "What term describes what is happening at point 'a' [the peak]?" Answer: **Depolarization**; main cause: **Na^+ influx through voltage-gated Na^+ channels.**

Exam Question: (2023 Q10 / 2024 Q10): "What term describes point 'b' [the downslope]?" Answer: **Repolarization**; main cause: **K^+ efflux through voltage-gated K^+ channels.**

Exam Question: (2024 Q32): "During depolarization, which ion channels open first?" Answer: **B) Sodium channels.**

4.4.3 Refractory Period

1. **Absolute refractory period** — neuron cannot fire at all. Na^+ channels are inactivated.
2. **Relative refractory period** — can fire, but only with a stronger stimulus. K^+ channels still partially open.

Exam Question: (2023 Q11 / 2024 Q11): "There are two periods labelled 'i' and 'ii'. What are they called?" Answer: i = **Absolute refractory period**; ii = **Relative refractory period.**

4.4.4 Propagation of the Action Potential

1. **Contiguous conduction** — continuous along unmyelinated axons. **Slow**.
2. **Saltatory conduction** — "jumps" between nodes of Ranvier in myelinated axons. **Much faster** and more energy-efficient.

Exam Question: (2024 Q36): "Saltatory conduction occurs in:" Answer: **B) Myelinated axons.**

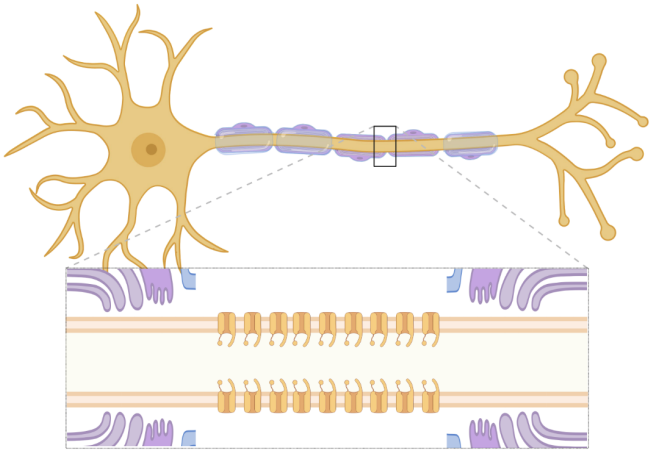


Figure 9: Contiguous vs saltatory conduction along unmyelinated and myelinated axons.

4.4.5 Properties of Action Potentials

a. **All-or-nothing**

b. **Non-decrementing** (does not weaken with distance)

c. **Unidirectional** (absolute refractory period behind)

d. Speed depends on axon diameter (larger = faster) and myelination (myelinated = faster)

Exam Question: (2023 Q11): "Which factor would increase the speed of nerve impulses?" Answer: **c) Increasing axon diameter because there is a greater surface area.**

4.4.6 Oligodendrocytes vs Schwann Cells ★

Both produce **myelin**, but in different locations:

a. **Schwann cells** — produce myelin in the **PNS** (peripheral nervous system). Each Schwann cell wraps around **one** segment of **one** axon, which provide insulation for speeding up electrical signal transmission .

b. **Oligodendrocytes** — produce myelin in the **CNS** (brain and spinal cord). Each oligodendrocyte can wrap around segments of **multiple** axons.

Exam Question: (2024 Q6): "Which type of cell in the CNS creates myelin sheaths around axons?" Answer: **C) Oligodendrocytes.**

Exam Question: (2024 Q14): "What is the role of myelin sheath in neurons?" Answer: **C) Speeding up signal transmission.**

Exam Question: (2025 MC5): "An Alpha fiber axon is to Schwann Cell as a coffee cup is to:" Answer: **b) The corrugated cardboard that forms a sleeve.**

4.5 Synapse ★

4.5.1 Main Components of a Synapse

a. Axon terminal — contains vesicles filled with neurotransmitter

b. Presynaptic neuron — sends the signal

c. Synaptic cleft — small gap between neurons

d. Postsynaptic cell — receives the signal

e. Postsynaptic receptors — bind neurotransmitters, open ion channels

f. Postsynaptic potential — EPSP or IPSP

Exam Question: (2024 Q24): "The synaptic cleft is the gap between:" Answer: **B) A presynaptic neuron and a postsynaptic neuron in a chemical synapse.**

4.5.2 Types of Synapses

1. **Chemical Synapse** — uses neurotransmitters across synaptic cleft
2. **Electrical Synapse** — uses gap junctions for direct ion flow. Very fast, no delay.

Step-by-step chemical synapse process:

1. AP arrives at axon terminal
2. Voltage-gated Ca^{2+} channels open → Ca^{2+} rushes in
3. Ca^{2+} causes vesicle fusion (exocytosis) → neurotransmitter released into synaptic cleft
4. Neurotransmitter diffuses across synaptic cleft
5. Neurotransmitter binds to postsynaptic receptors → ion channels open

6. Postsynaptic potential created (EPSP or IPSP)

Exam Question: (2024 Q21): "What happens when an action potential reaches the presynaptic terminal?" Answer: **A) It triggers the release of neurotransmitters.**

Exam Question: (2024 Q40): "Which ion's influx is crucial for the release of neurotransmitters?" Answer: **C) Calcium (Ca^{2+}).**

Exam Question: (2024 Q41): "Neurotransmitters are released into the synaptic cleft by:" Answer: **B) Axon terminals.**

Exam Question: (2025 MC6): "What is the most immediate action missing in Cytosis when you perform exocytosis?" Answer: **a) Ca^{2+} because it is needed for proteins to fuse vesicles with the plasma membrane.**

4.5.3 Molecular Transport at the Synapse ★

The 2023 and 2025 exams both ask you to label transport mechanisms in a synapse diagram (Figure 2 / Figure 2 style diagram). Know all four:

Label	Transport Type	Notes
a	Facilitated diffusion, channel-mediated	Ion channel in presynaptic membrane
b	Exocytosis	Vesicle fusion; NOT facilitated (needs ATP)
c	Facilitated diffusion, channel-mediated	Postsynaptic ionotropic receptor; ligand-gated
d	Secondary active transport, carrier-mediated	Reuptake of neurotransmitter into presynaptic termi

Exam trap: Exocytosis is **not** facilitated transport — it is an active process that requires ATP. Facilitated diffusion does not require ATP. Know the distinction.

Exam Question: (2025 Q13 / 2023 Q16–Q18): "Label the transport mechanisms at the synapse." Answer: see table above. "d" = secondary active transport / reuptake.

Exam Question: (2025 Q14 / 2023 Q17): "List two more common terms for the postsynaptic receptor (structure 'c')." Acceptable: **non-specific cation channel, ionotropic receptor, cholinergic receptor, ligand-gated ion channel.**

4.5.4 How the Synaptic Signal Ends ★

1. **Reuptake** — presynaptic terminal recaptures neurotransmitter via transporter proteins (carrier-mediated, secondary active transport). This is label "d" in the synapse diagram.
2. **Enzymatic degradation** — enzymes break the neurotransmitter apart in the cleft (e.g., **acetylcholinesterase** breaks down ACh).
3. **Diffusion** — neurotransmitter drifts away from the synapse.

Exam Question: (2023 Q18): "The circled area labelled 'd' shows one mechanism [reuptake]. List two additional processes that eliminate neurotransmitters from the synaptic cleft." Answer: **Enzymatic degradation** and **Diffusion.**

4.5.5 EPSP vs. IPSP ★

Property	EPSP	IPSP
Direction	Depolarizing (membrane potential rises)	Hyperpolarizing (membrane potential falls)
Main ion	Na^+ enters	Cl^- enters / K^+ exits
Effect	Increases likelihood of AP	Decreases likelihood of AP
All-or-nothing?	No — graded	No — graded

The **Grand PostSynaptic Potential (GPSP)** is the **algebraic sum** of all EPSPs and IPSPs arriving at the axon hillock at a given moment. It is not a separate type of potential — it is the net result of summation.

Exam Question: (2024 Q22): "EPSP increases the likelihood of:" Answer: **A) An action potential.**

Exam Question: (2024 Q7): "Hyperpolarization of a neuron makes it:" Answer: **A) Less likely to fire an action potential.**

Exam Question: (2025 MC15): "The net summation brings membrane potential to –60 mV. Will an action potential occur?" Answer: **a) No, an action potential will not occur** (threshold is –55 mV; –60 mV is below threshold).

4.5.6 GABA and Glutamate ★

GABA (γ-aminobutyric acid) is the main **inhibitory** neurotransmitter in the brain.

- $GABA_A$ receptor (ionotropic) → Cl^- channels open → Cl^- moves inward → hyperpolarization
- $GABA_B$ receptor (metabotropic/GPCR) → K^+ channels open → K^+ moves outward → hyperpolarization
- Both produce an **IPSP** and make the neuron less likely to fire

Glutamate is the main **excitatory** neurotransmitter in the brain. Glutamate binds → Na^+ channels open → depolarization (EPSP).

Glycine is an **inhibitory** neurotransmitter in the spinal cord and brainstem.

Exam Question: (2025 Q6): "What will happen to the membrane potential of an old postsynaptic neuron if a presynaptic neuron releases GABA?" Answer: GABA binds $GABA_A$ receptor (ligand-gated); Cl^- moves inward; postsynaptic neuron **hyperpolarizes**. Key values: $V_m = -72.5$ mV; $E_{Cl^-} = -77.6$ mV.

Exam Question: (2023 Q14): Same question framed for a young neuron. Answer: GABA opens Cl^- channels; Cl^- moves inward; **hyperpolarization** (IPSP). $V_m = -72.5$ mV; E_{Cl^-} calculated from young neuron concentrations.

4.5.7 Receptor Types ★

Receptors are named after the **neurotransmitter** they respond to:

- **Cholinergic** receptors → respond to **Acetylcholine (ACh)**
- **Adrenergic** receptors → respond to **Adrenaline / Norepinephrine**
- **Dopaminergic** receptors → respond to **Dopamine**
- **GABAergic** receptors → respond to **GABA**

The suffix **-ergic** means "activated by" that neurotransmitter.

Every receptor falls into one of two categories based on *how* it signals:

1. **Ionotropic receptor** (ligand-gated ion channel)
   - The receptor **IS** an ion channel
   - Neurotransmitter binds → channel opens immediately → ions flow
   - **Fast** (milliseconds), **short-lasting**
2. **Metabotropic receptor** (G-protein coupled receptor / GPCR)
   - The receptor is **NOT** an ion channel
   - Neurotransmitter binds → activates G-protein → second messenger cascade
   - **Slower** (seconds–minutes), **longer-lasting**

Feature	Ionotropic	Metabotropic
Other name	Ligand-gated ion channel	GPCR
Mechanism	Receptor IS the channel	Activates G-protein
Speed	Fast (ms)	Slow (sec–min)
Duration	Short	Long
Pathway	Direct ion flow	Second messenger cascade

Acetylcholine (ACh) and Its Receptors ★

Acetylcholine (ACh) is used at:

- The **neuromuscular junction** (skeletal muscle contraction)
- The **parasympathetic nervous system** ("rest and digest")
- Parts of the **brain** (memory, attention)

Both of its receptors are **cholinergic** (NOT adrenergic — common exam trap):

1. **Nicotinic receptor** — named after nicotine
   - Type: **Ionotropic** (ligand-gated ion channel)
   - ACh binds → Na^+ rushes in → depolarization (fast)
   - Location: **neuromuscular junction** (N1), **all autonomic ganglia** (N2)
2. **Muscarinic receptor** — named after muscarine (mushroom toxin)

- Type: **Metabotropic** (GPCR)
- ACh binds → G-protein → second messenger cascade (slower)
- Location: **parasympathetic target organs**

Feature	Nicotinic	Muscarinic
Named after	Nicotine (cigarettes)	Muscarine (mushroom toxin)
Neurotransmitter	ACh	ACh
Receptor class	Cholinergic	Cholinergic
Receptor type	**Ionotropic** (ion channel)	**Metabotropic** (GPCR)
Mechanism	Na^+ influx → depolarization	G-protein → second messenger
Speed	Fast (milliseconds)	Slow (seconds–minutes)
Main location	NMJ + autonomic ganglia	Parasympathetic target organs

Common exam traps:

- "Nicotinic and muscarinic are **adrenergic** receptors" → **FALSE.** They are **cholinergic.**
- "Nicotinic receptors are metabotropic" → **FALSE.** Nicotinic = ionotropic; Muscarinic = metabotropic.
- Muscle at motor endplate is always activated by **ACh** via **nicotinic** receptors.

Exam Question: (2025 MC7): "Nicotinic and muscarinic receptors are two kinds of adrenergic receptors." Answer: **a) False, they are cholinergic receptors.**

Exam Question: (2024 Q9): "What neurotransmitter is always associated with muscle activation at the motor endplate?" Answer: **D) Acetylcholine.**

4.6 Synaptic Integration ★

A single neuron receives inputs from many neurons simultaneously. Each produces an EPSP or IPSP. The **axon hillock** sums all of these (the GPSP) and fires an action potential only if the net depolarization reaches threshold (–55 mV).

4.6.1 Temporal Summation

The **same** presynaptic neuron fires repeatedly in rapid succession. EPSPs accumulate before previous ones decay.

4.6.2 Spatial Summation

Multiple different presynaptic neurons fire simultaneously, each contacting the same postsynaptic neuron. Their EPSPs add together at the axon hillock.

Exam Question: (2025 Q10 / 2024 Q10 / 2023 Q13): "What could cause the different responses [in the neuronal integration figure]?" Answer: **Temporal summation** and **Spatial summation.** — Tested **every year.**

Exam Question: (2024 Q47): "The Grand Post-Synaptic Potential (GPSP) can be affected by:" Answer: **A) Temporal summation.**

Exam Question: (2025 MC15): "Net summation brings membrane potential to –60 mV. Will an action potential occur?" Answer: **a) No** — –60 mV does not reach the threshold of –55 mV.

4.6.3 Lateral Inhibition ★

An excited neuron **inhibits its neighbours** to sharpen sensory perception and increase contrast. The most strongly stimulated neuron suppresses weaker adjacent signals → enhanced spatial resolution.

Exam Question: (2024 Q27): "What mechanism enhances the precision of sensory perception by reducing the activity of neighboring neurons?" Answer: **C) Lateral inhibition.**

4.6.4 Presynaptic Inhibition

An **inhibitory neuron synapses directly onto a presynaptic terminal**, reducing how much neurotransmitter that terminal releases. Different from postsynaptic inhibition (IPSP), which acts on the postsynaptic cell.

Exam Question: (2024 Q37): "Presynaptic inhibition occurs when:" Answer: **D) An inhibitory neuron synapses directly with a presynaptic neuron.**

5 Central Nervous System (CNS)

The **CNS** consists of the brain and spinal cord — the integration hub of the body.

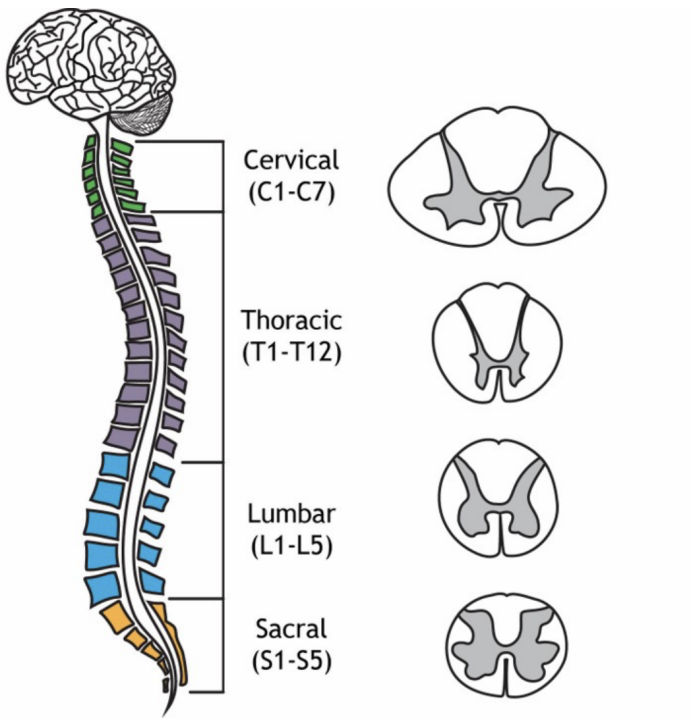


Figure 3: CNS.

5.1 Grey vs White Matter

- **Grey matter** — neuronal cell bodies. Site of synaptic integration.
- **White matter** — myelinated axons. Signal transmission highways.

5.2 Spinal Cord ★

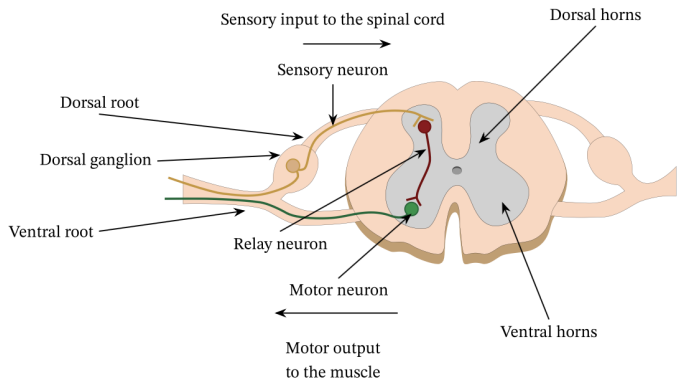


Figure 3: spine cross section, relay neuron is what we call inhibitory interneuron

Structure	Function	Contains
Dorsal horn	Sensory processing	Interneurons receiving sensory input
Ventral horn	Motor output	Motor neuron cell bodies
Dorsal root	Afferent (sensory) fibres enter	Cell bodies in dorsal root ganglion
Ventral root	Efferent (motor) fibres exit	Axons of motor neurons

Bell–Magendie Law: Dorsal = sensory in; Ventral = motor out.

Exam Question: (2024 Q46): "Which part of the spinal cord contains motor neurons?" Answer: **C) Ventral horn.**

5.3 Ascending vs Descending Tracts

- **Ascending tracts** — carry **sensory** information **to** the brain (afferent)
- **Descending tracts** — carry **motor** commands **from** the brain (efferent)
- White matter contains **both** ascending and descending tracts

Exam Question: (2023 Q24): "Number 6 points to a region in the spinal cord. Does it have ascending, descending, or no tracts?" Answer: **Both ascending and descending tracts** — it is white matter, which contains both.(the diagram is unclear, if its ventral horn, then it's descending

Exam Question: (2025 Q17): "The postsynaptic neuron (number 5) travels from this level of the spinal cord to the brain. What is this pathway called?" Answer: **Ascending tract.** Also accepted: Sensory pathway, Afferent pathway, Somatosensory pathway. NOT accepted: Afferent fiber.

A **reflex** is a fast, involuntary, stereotyped response processed at the spinal cord level — bypasses the brain for speed.

5.4 Reflexes ★

A **reflex** is a fast, involuntary, stereotyped response processed at the spinal cord level — bypasses the brain for speed.

5.4.1 Reflex Arc — The Five Components

1. **Receptor** — senses the stimulus
2. **Afferent (sensory) neuron** — carries signal to spinal cord via **dorsal root**
3. **Integration centre** (spinal cord interneurons) — processes the signal
4. **Efferent (motor) neuron** — carries motor command via **ventral root**
5. **Effector** — muscle or gland that produces the response

Exam Question: (2024 Q48): "Reflex arcs do NOT include which component?" Answer: **D) Hormone.**

Exam Question: (2023 Q19): "What type of receptor is shown in number 1 [in spinal reflex diagram]?" Answer: Mechanoreceptor (or sensory receptor).

Exam Question: (2023 Q20): "What kind of neuron is labelled number 2?" Answer: Afferent neuron / sensory neuron.

5.4.2 Monosynaptic vs Polysynaptic Reflexes ★

- **Monosynaptic reflex** — **one synapse** between sensory and motor neuron (no interneuron). Example: **knee-jerk (patellar) reflex**. Fastest possible.
- **Polysynaptic reflex** — **two or more synapses**; involves interneurons. Example: **withdrawal reflex** (hot stove). Slower but more complex.
- Reflex arcs can be **both**.

Exam Question: (2024 Q49): "Reflex arcs may be:" Answer: **D) Both A and B** (both monosynaptic and polysynaptic).

Exam Question: (2023 Q12): "A monosynaptic reflex compared to an email exchange is most like:" Answer: **c) An automatic reply email.**

Exam Question: (2025 MC10): "Motor neurons create the fastest reflexes because they:" Answer: **c) Have large diameter axons and target skeletal muscles have nicotinic receptors.**

5.4.3 Withdrawal and Crossed Extensor Reflexes

- **Withdrawal reflex**: painful stimulus → flexor contracts, pulling limb away. Interneurons inhibit antagonist extensors (**reciprocal inhibition**).
- **Crossed extensor reflex**: accompanies withdrawal to maintain balance. While stimulated limb flexes, the opposite limb extends.

Feature	Withdrawal Reflex	Crossed Extensor Reflex	Why
Which limb	Stimulated limb	**Opposite** limb	Works together simultaneously
Movement	**Flexion** (pulls away)	**Extension** (pushes down)	Opposite movements
Purpose	Remove limb from danger	Maintain balance	Prevents falling
Type	Polysynaptic	Polysynaptic	Involves interneurons

One line to remember: Same side **flexes**, opposite side **extends** — always simultaneously.

How this could be tested — likely scenario question:
"You step on a nail with your right foot. Describe what happens to both legs."

- **Right leg** (stimulated): withdrawal reflex → flexor contracts, pulls right foot up and away. Interneurons inhibit extensor via reciprocal inhibition.
- **Left leg** (opposite): crossed extensor reflex → left leg extends to support body weight so you don't fall.

Exam Question: Likely exam trap: "The crossed extensor reflex occurs in the same limb as the withdrawal reflex." Answer: **False** — it occurs in the **opposite** limb.

Exam Question: Likely exam question: "What is the purpose of the crossed extensor reflex?" Answer: **To maintain balance** when one limb is pulled away from a stimulus.

5.4.4 Reflex Arc — The Five Components

1. **Receptor** — senses the stimulus
2. **Afferent (sensory) neuron** — carries signal to spinal cord via **dorsal root**
3. **Integration centre** (spinal cord interneurons) — processes the signal
4. **Efferent (motor) neuron** — carries motor command via **ventral root**
5. **Effector** — muscle or gland that produces the response

Exam Question: (2024 Q48): "Reflex arcs do NOT include which component?" Answer: **D) Hormone.**

Exam Question: (2023 Q19): "What type of receptor is shown in number 1 [in spinal reflex diagram]?" Answer: Mechanoreceptor (or sensory receptor).

Exam Question: (2023 Q20): "What kind of neuron is labelled number 2?" Answer: Afferent neuron / sensory neuron.

5.5 The Spinal Reflex Arc Diagram ★

A **reflex** is a fast, involuntary, stereotyped response processed at the spinal cord level — it bypasses the brain for speed. This diagram is tested **every year** — 2023 had 6 questions on it, 2025 had 3.

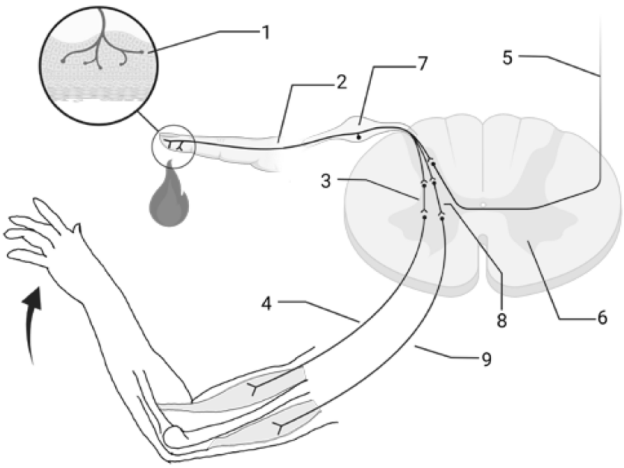


Figure 3: Spinal reflex arc. Numbers 1–9 are all exam targets.

What Each Label Is

Label	Structure	State	Why
1	Mechanoreceptor in fingertip	Excited	Stimulus (touch/pain) activates it
2	Afferent (sensory) neuron	Excited	Carries signal inward toward spinal cord
3	Interneurons inside spinal cord	Excited	Receives and relays the sensory signal
4	Efferent (motor) neuron	Excited	Carries motor command out to flexor muscle
5	Ascending tract neuron	Excited	Carries signal up to brain (somatosensory cortex) for conscious awareness
6	Ventral horn	Excited	Contains motor neuron cell bodies; part of efferent pathway
7	Dorsal root ganglion	Excited	Cell bodies of sensory neurons cluster here
8	Inhibitory interneuron	**Excited**	Excited so it CAN inhibit neuron 9
9	Antagonist motor neuron	**Inhibited**	Inhibited by neuron 8 (reciprocal inhibition)

Key Concepts to Understand the Diagram

Interneuron vs Inhibitory Interneuron

An **interneuron** is any neuron that sits *between* the sensory and motor neuron inside the spinal cord — it is just a relay/processing middleman. An **inhibitory interneuron** is a specific type whose job is to **suppress** another neuron (thats connecting to it instead of beside it) by releasing inhibitory neurotransmitters (GABA or glycine).

- All inhibitory interneurons are interneurons
- Not all interneurons are inhibitory — some are excitatory (like neuron 3, which passes the signal forward)
- You know neuron 8 is inhibitory because it connects to neuron 9 and its job is to shut it down

Most common exam trap — label 8: The inhibitory interneuron is itself **excited**, NOT inhibited. It must be excited in order to do its job of inhibiting neuron 9. An inhibitory neuron that is itself inhibited would do nothing.

What is the Antagonist?

Every muscle has an opposite muscle:

Type	What it does	In this reflex
Flexor (agonist)	Decreases joint angle — bends/pulls inward	Pulls arm away from stimulus
Extensor (antagonist)	Increases joint angle — straightens outward	Would push arm back toward stimul

The stimulus is pain on the finger → natural response is to **flex** (pull away). The flexor must contract (neuron 4 excited) and the extensor must relax (neuron 9 inhibited) simultaneously. If both contracted at the same time you would not move at all — they would fight each other. This is why **reciprocal inhibition** exists.

How do you know which motor neuron is agonist vs antagonist?

- Whichever motor neuron is **excited** = agonist (doing the movement)
- Whichever motor neuron is **inhibited** = antagonist (being silenced)
- Neuron 8 (inhibitory interneuron) always connects to the antagonist — whatever neuron 8 synapses onto = antagonist motor neuron

For this diagram specifically: Label 1 is a **mechanoreceptor** because the stimulus shown is physical touch/pressure on the fingertip. However if the question changes the scenario to pain → answer is **nociceptor**. If heat → answer is **thermoreceptor**. Always read the scenario carefully.

The Full Signal Flow Step by Step

1. Stimulus hits finger → **receptor (1)** activated
2. **Afferent neuron (2)** carries signal inward via dorsal root
3. Signal passes through **dorsal root ganglion (7)** where cell bodies sit
4. Inside spinal cord, signal splits **three ways simultaneously**:
   a. → **interneuron (3)** → **efferent motor neuron (4)**: flexor muscle contracts — the actual reflex movement (arm pulls up, shown by arrow)
   b. → **inhibitory interneuron (8)**: inhibits antagonist motor neuron (9) so extensor relaxes and does not fight the flexion
   c. → **ascending tract neuron (5)**: carries signal up to somatosensory cortex in brain so you consciously feel it

Important: The reflex movement (step a) happens at the spinal cord level **before** the brain is even aware of it (step c). That is why reflexes are so fast — the signal does not need to travel all the way to the brain and back.

All Exam Questions on This Diagram

Exam Question: (2023 Q19): "What type of receptor is shown in number 1?" Answer: **Mechanoreceptor** (sensory receptor). It is in the fingertip responding to touch/pressure.

Exam Question: (2023 Q20): "What kind of neuron is labelled number 2?" Answer: **Afferent neuron** (also accepted: sensory neuron).

Exam Question: (2023 Q21): "Where does neuron 5 eventually project to in the brain to have awareness that the finger is being stimulated?" Answer: **Somatosensory cortex.**

Exam Question: (2023 Q22): "What does number 3 represent? Is it excited or inhibited?" Answer: **Interneuron (dorsal root ganglion also accepted). It is excited.**

Exam Question: (2023 Q23): "What does number 9 represent? Is it excited or inhibited?" Answer: **Antagonist motor neuron. It is inhibited** — neuron 8 suppresses it via reciprocal inhibition.

Exam Question: (2023 Q24): "Number 6 points to a region in the spinal cord. Does it have ascending, descending, or no tracts?" Answer: **Descending tracts** — label 6 is the ventral horn, which receives motor commands coming down from the brain.

Exam Question: (2025 Q15): "What is the best label for number 7?" Answer: **Dorsal root ganglion.**

Exam Question: (2025 Q16): "What kind of neuron is labelled number 8? Is it excited or inhibited?" Answer: **Inhibitory interneuron. It is excited.**

Exam Question: (2025 Q17): "Neuron 5 travels from this level of the spinal cord to the brain. What is this pathway called?" Answer: **Ascending tract.** Also accepted: Sensory pathway, Afferent pathway, Somatosensory pathway. NOT accepted: Afferent fiber.

Why Does Neuron 9 Get Inhibited? — Reciprocal Inhibition

When you touch something hot and pull your hand away, your **flexor** muscle contracts to pull the limb in. At the same time, the **extensor** (antagonist) muscle must relax — otherwise the two muscles would fight each other and you would not move efficiently.

Neuron 8 (inhibitory interneuron) is responsible for this. It receives the same sensory signal and simultaneously sends inhibitory signals to the extensor motor neuron (9), shutting it down. This is called **reciprocal inhibition**.

Muscle	Motor neuron	What happens
Flexor	Neuron 4	**Excited** — contracts, pulls limb away
Extensor (antagonist)	Neuron 9	**Inhibited** — relaxes, allows flexion

The Full Signal Flow Through the Diagram

Tracing the pathway step by step:

1. Stimulus hits finger → **receptor (1)** activated
2. **Afferent neuron (2)** carries signal inward via dorsal root
3. Signal passes through **dorsal root ganglion (3/7)** where cell bodies sit
4. Inside spinal cord, signal splits two ways simultaneously:
    a. → **efferent motor neuron (4)**: triggers flexor muscle contraction (the reflex)

   b. → **inhibitory interneuron (8)**: inhibits antagonist motor neuron (9)
   c. → **ascending tract neuron (5)**: carries signal up to somatosensory cortex so you consciously feel it

5. **White matter (6)** contains the ascending tracts carrying signal toward brain

Exam Question: (2023 Q24): "Number 6 points to a region in the spinal cord. Does this area have ascending, descending, or no tracts?" Answer: **Both ascending and descending tracts.** White matter contains both. Explanation: white matter surrounds the grey matter butterfly and carries signals in both directions.

6 Nervous System Overview & Reflexes ★

6.1 Divisions of the Nervous System

1. **Central Nervous System (CNS)** — brain + spinal cord
2. **Peripheral Nervous System (PNS)** — all nerves outside the CNS
   a. **Afferent (sensory)** division — carries signals TO the CNS
   b. **Efferent (motor)** division — carries signals FROM the CNS
      - Somatic NS → skeletal muscle (voluntary)
      - Autonomic NS → smooth muscle, cardiac muscle, glands (involuntary)

Exam Question: (2024 Q15): "The brain and spinal cord are part of the:" Answer: **B) Central Nervous System.**

Exam Question: (2024 Q12 / 2025 MC12): "What is the function of afferent neurons?" Answer: **A) To transmit messages from receptors to the spinal cord and brain.**

Exam Question: (2024 Q50): Mnemonic: **SAME** — Sensory Afferent, Motor Efferent.

6.2 Receptor Physiology ★

Sensory receptors convert stimuli into electrical signals. The CNS extracts **four key attributes** from every sensory signal: **Modality**, **Intensity**, **Locality**, and **Duration**.

6.2.1 Receptor (Generator) Potential

When a receptor is stimulated, it produces a **receptor potential** — a local, graded depolarization.

- **Graded** — amplitude proportional to stimulus strength; NOT all-or-nothing
- If large enough, it triggers action potentials in the afferent neuron

Exam Question: (2023 Q13): "Which statement applies to receptor potentials?" Answer: **a) They are graded potentials.**

6.2.2 Receptor Types by Adequate Stimulus

a. **Mechanoreceptors** — mechanical deformation (touch, pressure, vibration, stretch)

b. **Thermoreceptors** — temperature changes

c. **Nociceptors** — pain / tissue damage

d. **Photoreceptors** — light

e. **Chemoreceptors** — chemicals (taste, smell, blood O_2)

Exam Question: (2024 Q10): "Ion channels that open and close in response to mechanical loading are associated with:" Answer: **A) Mechanoreceptors.**

6.2.3 1. Modality — "What kind of stimulus?"

Determined by **which dedicated pathway** to the brain is activated (**Labeled Line Principle**), not by the signal itself.

6.2.4 2. Intensity — "How strong?" ★

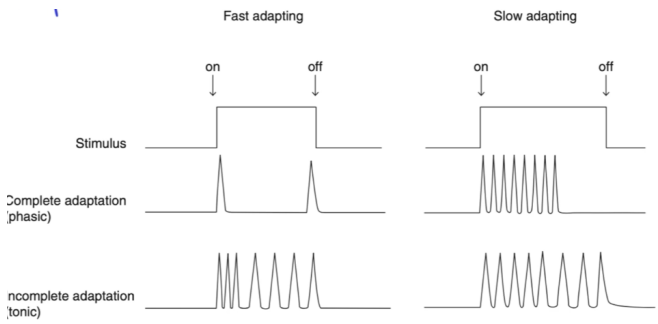
Encoded by **frequency coding**: stronger stimulus → larger receptor potential → **higher frequency** of APs. AP size does NOT change — only frequency.

Exam Question: (2024 Q29): "The larger the receptor potential, the:" Answer: **B) Higher the frequency of action potentials.**

6.2.5 3. Locality — "Where?"

Determined by **receptive fields**. Smaller receptive field = better spatial resolution (e.g., fingertips vs. back). **Two-point discrimination** measures this. Lateral inhibition sharpens locality further.

6.2.6 4. Duration — "How long?" ★



Sonic and Phasic receptor

Type	Also called	Adaptation	Detects
Rapidly adapting	**Phasic**	Stops responding quickly	Changes (on/off)
Slowly adapting	**Tonic**	Keeps firing throughout	Ongoing stimuli

Examples: Phasic = Pacinian corpuscle (vibration), Meissner's (light touch). Tonic = Merkel cells (pressure), nociceptors (pain).

Exam Question: (2024 Q19 / 2024 Q28 / 2023 Q15): "What type of receptor adapts rapidly?" Answer: **Phasic receptor.**

Exam Question: (2023 Q16): "Which sensory receptor neuron exhibits tonic adaptation?" Answer: **Both the top and bottom receptors** (those that keep firing during the entire stimulus duration).

6.3 Somatic Nervous System (SNS)

Controls voluntary movement of skeletal muscle. **One neuron** runs directly from CNS to muscle.

- Neurotransmitter: **Acetylcholine (ACh)**
- Receptor on muscle: **Nicotinic** (ionotropic, fast)
- Effect: Always **excitatory**

Exam Question: (2025 Q18): "What type of neuron is shown in row a?" Answer: **Somatic Neuron** (also accepted: Motor neuron).

7 Autonomic Nervous System (ANS) ★

The **ANS** controls involuntary functions: smooth muscle, cardiac muscle, and glands.

Key structural difference from somatic: The ANS uses **two neurons in series** from CNS to target organ.

1. **Preganglionic neuron** — cell body in CNS; axon travels to a ganglion
2. **Postganglionic neuron** — cell body in ganglion; axon travels to target organ

Exam Question: (2024 Q11): "The autonomic nervous system typically has a chain of how many neurons from the CNS to the effector organ?" Answer: **B) Two.**

Exam Question: (2024 Q8): "The autonomic nervous system is divided into which two primary systems?" Answer: **B) Sympathetic and Parasympathetic.**

Exam Question: (2025 MC1): "In the transit analogy (airport → bus terminal → home), what represents the place where they wait for the bus?" Answer: **a) The chain ganglion.**

7.1 Sympathetic Division ("Fight or Flight")

Also called the **thoracolumbar** division (preganglionic neurons originate from **T1–L2**).

- Increases heart rate and blood pressure
- **Dilates** pupils (widens to see threats)
- Dilates airways (bronchodilation)
- Reduces digestive activity

- Postganglionic neurotransmitter: **Norepinephrine (NE)** → **adrenergic receptors**
- Fibre length: **short preganglionic, long postganglionic**

The Adrenal Medulla: A modified sympathetic ganglion on top of each kidney. Preganglionic neuron synapses directly onto **chromaffin cells**, which release **epinephrine** (∼80%) and **norepinephrine** (∼20%) into the bloodstream — amplifying fight-or-flight body-wide.

Exam Question: (2023 Q20): "Which spinal nerves do sympathetic fibres emanate from?" Answer: **a) Thoracic and lumbar levels.**

7.2 Parasympathetic Division ("Rest and Digest")

Also called the **craniosacral** division (preganglionic neurons from **brainstem cranial nerves III, VII, IX, X** and **sacral S2–S4**).

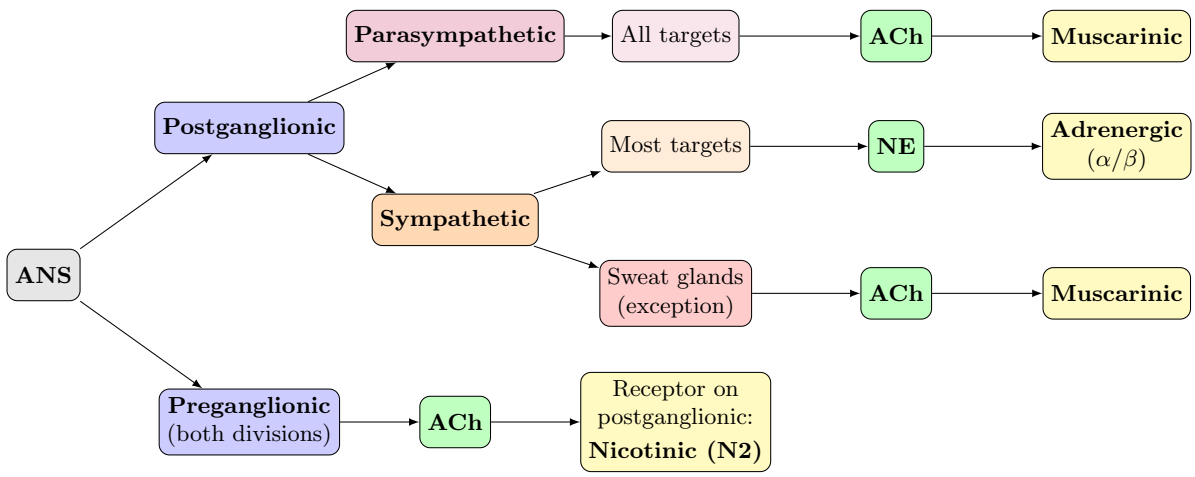
- Decreases heart rate
- Stimulates digestion
- Constricts pupils
- Postganglionic neurotransmitter: **Acetylcholine (ACh)** → **muscarinic receptors**
- Fibre length: **long preganglionic, short postganglionic**

Exam Question: (2024 Q45): "Which type of receptors are associated with the parasympathetic nervous system?" Answer: **D) Cholinergic.**

Exam Question: (2024 Q20): "Adrenergic receptors in the ANS are activated by:" Answer: **A) Norepinephrine.**

Exam Question: (2023 Q17): "Which statement best describes parasympathetic postganglionic fibres?" Answer: **c) They are cholinergic fibres.**

7.3 ANS Neurotransmitter and Receptor Summary ★



Neuron	NT Released	Receptor on Target	Type
All preganglionic (symp + para)	ACh	Nicotinic (N2)	Ionotropic
Parasympathetic postganglionic	ACh	Muscarinic	Metabotropic
Sympathetic postganglionic (most)	NE	Adrenergic (α/β)	Metabotropic
Sympathetic postganglionic (sweat glands)	ACh	Muscarinic	Metabotropic
Somatic motor neuron	ACh	Nicotinic (N1)	Ionotropic

Critical rule: ALL preganglionic neurons (both sympathetic and parasympathetic) release **ACh** onto **nicotinic receptors** on postganglionic cell bodies. Therefore **all postganglionic neurons have nicotinic receptors on their dendrites/cell bodies.**

Exam Question: (2025 MC13 / 2023 Q18): "All postganglionic neurons have ____ on their dendrites or cell bodies." Answer: **a) Nicotinic receptors.**

Exam Question: (2023 Q19): "Which best describes nicotinic receptors?" Answer: **b) They respond to ACh released from both sympathetic and parasympathetic preganglionic fibres.**

Exception: Sympathetic Cholinergic Fibres

Most sympathetic postganglionic fibres are adrenergic (NE). **Exception:** some sympathetic postganglionic fibres to **sweat glands** release **ACh** onto **muscarinic** receptors.

Exam Question: (2025 Q20): "Row d has qualities of both row b and c. How is this nerve the same but different?" Answer: **Sympathetic nerve that releases acetylcholine (ACh).**

Exam Question: (2025 Q19): "What type of receptor is attached to the effector cell in row b?" Answer: **Muscarinic receptor** (cholinergic).

7.4 Somatic vs. Autonomic Comparison ★

Feature	Somatic	Autonomic
Neurons in chain	1	2 (pre- and postganglionic)
Target	Skeletal muscle	Smooth/cardiac muscle, glands
Control	Voluntary	Involuntary
NT at target	ACh (nicotinic N1)	ACh (muscarinic) or NE (adrenergic)
Effect	Always excitatory	Can excite **or** inhibit

7.5 Structure of a Peripheral Nerve

Memory trick (inside → out): **Every Person Eats** — Endoneurium → Perineurium → Epineurium.

Layer	**What it wraps**	**Position**
Endoneurium	Individual axon (+ myelin)	Innermost
Perineurium	One fascicle (bundle of axons)	Middle
Epineurium	Entire nerve (all fascicles)	Outermost